

Figure 5: Creating a new file in Python 3 IDLE

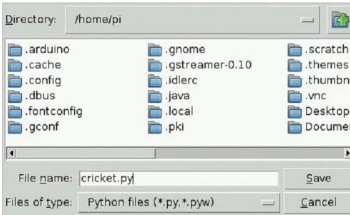


Figure 6: Saving a Python file

```
print("Error in retrieving the current cricket matches")
```

Notice the following points about the program code:

1. We are using the requests and HTML package. So ensure that you have used Pip to install the packages.
2. We invoke the `get` method from the requests package, which makes an HTTP GET call to the cricket API. You can paste the URL <http://cricapi.com/api/cricket> in any browser to see the JSON data that is returned.
3. If the HTTP response returned is 200, which means OK, then we are using the in-built JSON decoder available in the requests library to extract the data element, which is the root element.
4. We finally iterate through all the records, i.e., matches that are currently happening and then print out the title attribute, which contains the team names and the scores.

To run the program from IDLE, you can simply press F5. This will prompt you to save the file and you can save it with a name of your choice. We choose *cricket.py* and it is saved in the */home/pi* folder.

The program will then be executed in the Python IDLE shell and you will see the output with the list of matches.

Another way of running the application is from the terminal. Simply launch the terminal and go to the */home/pi* folder where you saved the *cricket.py* file. Next, execute the application using the Python 3 interpreter as shown below:

```
$ python3 cricket.py
England 297/10 & 414/5 * v Pakistan 490/10
Essex 358/10 & 163/4 * v Sussex 448/10
```

```
Hampshire 548/6 v Lancashire 310/4 *
Leicestershire 380/10 & 109/5 * v Derbyshire 362/10
Middlesex 293/10 v Surrey 415/10 & 234/6 *
Australia A 181/1 * v South Africa A 304/10
Zimbabwe v New Zealand 436/4 *
South Africa Emerging Players v Sri Lanka Development
Emerging Team 129/2 *
Northern Knights v North-West Warriors
Western Storm v Surrey Stars
Guyana Amazon Warriors v Jamaica Tallawahs
```

GPIO programming

The Raspberry Pi is often used in conjunction with other hardware to create interesting electronic projects. The Pi 3 comes with 40 GPIO pins that you can use to interface with various hardware devices—for both receiving data from them or for writing data to them. This way, we can write applications to both read and also to control devices, i.e., turn them on and off, etc.

To program the GPIO pins on Raspberry Pi with Python, there is a Raspberry Pi GPIO Python library that is made available. The following command is used to install the GPIO library for Python 3:

```
sudo apt-get install python3-rpi.gpio
```

Once the library is installed, we can use it in a simple Python application, the quintessential LED blinking program as shown below (remember to launch the IDLE shell or the Python shell at the terminal with `sudo` privileges). This assumes that you have connected a breadboard, LED, resistor to the Raspberry Pi GPIO and the LED is being driven via pin 18.

```
import RPi.GPIO as GPIO
import time
GPIO.setmode(GPIO.BCM) # set board mode to Broadcom
GPIO.setup(18, GPIO.OUT) # set up pin 18
```

while True:

```
    GPIO.output(18, 1) # turn on pin 18
    time.delay(1) #delay for 1 seconds
    GPIO.output(18,0) #turn off pin 18
```



References

- [1] Raspberry Pi Home: www.raspberrypi.org
- [2] Raspberry Pi Hardware (<https://www.raspberrypi.org/learning/hardware-guide/components/raspberry-pi/>)
- [3] Python: www.python.org

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